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Development of AI-tools for making sense of future complex intelligent systems

A case study of the development of an AI- tool for planning energy systems

Elinor Särner



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“The best way to predict the future, is to create it.”

Drucker (2009)

Abstract

Artificial intelligence (AI) is increasingly introduced into many systems that modern society rely on and is often portrayed as a savior that can contribute to finding solutions to societal challenges e.g., social, and ecological sustainability. Many of these systems can be classified as complex systems, with interdependencies, emergent behaviors and a diversity of actors involved. As AI is increasingly introduced into these systems, we witness a transformation from complex systems into complex intelligent systems. At the same time caution is invoked toward the risks of AI regarding e.g., biases and loss of control as more tasks are transferred to AI. Hence, the introduction of AI into complex systems is associated with uncertainties around management of AI initiatives and their influence on future systems. Challenges like this can affect many different functions and professions and thus need to be understood collectively.

The aim of this thesis is to examine how the actors' prospective collective sensemaking processes in developing complex systems are affected by AI introduction. Previous research within complex systems literature shows important aspects regarding sensemaking of the system and situations within operations of complex systems. However, sensemaking in the development process of complex systems has been less studied. By examining the introduction of AI in complex systems development this thesis explores collective prospective sensemaking processes in the development of complex intelligent systems.

To study an emerging phenomenon like AI introduction in complex systems, an explorative case study was found suitable. The case chosen for the study was a cross-organizational development project of an AI-tool based on Machine Learning for planning of energy systems to be used in the urban planning process of new city districts. This setting revealed plenty prospective and collective sensemaking occasions around AI introduction and exhibited continuous engagement in prospective collective sensemaking relating to the development of the AI tool and the imagined use of the AI tool in the system, which have been reported in the two appended papers. The first paper showed misalignment between actors' sensemaking processes that alternated between seeking and disengaging behaviors. It also identified the use of boundary objects to retain disengaged actors and raised considerations around the level of detail of the boundary objects in relation to the sensemaking behaviors. The second paper identified dependencies between near- and distant-sensemaking loops that highlight challenges to connect retrospective insights with prospective imaginations by action in the present. In the creation of complex intelligent systems, human involvement seems inevitable, and the second paper exposes how AI can augment human cognition and organizational capabilities for creative imagination around possible and desirable distant-future scenarios.

This thesis extends previous research on prospective and collective sensemaking in the development process of complex intelligent systems by presenting a framework of near and distant future sensemaking and internal and external complexity. This provided new insights of how knowledge flows over system levels and how to use boundary objects throughout such projects. Insight that can be useful for management of purposeful AI introduction in complex systems and society. Moreover, it contributes with an empirical case of AI introduction in complex systems to the innovation management literature.

Sammanfattning

Artificiell intelligens (AI) integreras alltmer i de system som vårt moderna samhälle vilar på och framställs ofta som en viktig faktor för att lösa de utmaningar som samhället står inför, till exempel social och ekologisk hållbarhet. Många av dessa system kan definieras som komplexa system, med ömsesidiga beroenden, oförutsedda beteenden och en mångfald av inblandade aktörer. När AI introduceras i sådana system kan vi skönja en transformation från komplexa system till komplexa *intelligenta* system. Samtidigt påtalas ofta riskerna med AI avseende till exempel partiskhet och minskad kontroll när uppgifter tas över av AI. Införandet av AI i samhällskritiska system förknippas därmed med osäkerheter kring hanteringen av AI-initiativ och dess påverkan på framtiden. Detta leder till utmaningar som berör många olika funktioner och professioner och därmed behöver förstås kollektivt.

Syftet med avhandlingen är att undersöka aktörernas framtidsorienterade, gemensamma, meningsskapande processer under utveckling av komplexa system och hur de påverkas av introduktionen av AI. Tidigare forskning inom komplexa system har visat på viktiga aspekter gällande människors förståelse av systemet och situationer framförallt inom driften av komplexa system. Meningsskapande i utvecklingsprocessen av komplexa system har dock hittills inte uppmärksammats i samma utsträckning. Genom att undersöka introduktionen av AI i utvecklingen av komplexa system utforskar denna avhandling kollektiva, framåtblickande, meningsskapande processer inom utvecklingen av komplexa intelligenta system.

För studier av ett framväxande fenomen som AI-introduktion i komplexa system ansågs en explorativ fallstudie lämplig. Det valda fallet var ett utvecklingsprojekt av ett AI-verktyg baserat på maskininlärning med syfte att användas i planeringen av energisystem inom stadsplaneringsprocessen av nya stadsdelar och hade flera medverkande organisationer. Fallet visade på flera framtidsorienterade och gemensamma meningsskapande situationer kring AI-introduktion. Därmed synliggjordes de medverkandes kontinuerliga deltagande i framåtblickande gemensamt meningsskapande relaterat till utvecklingen av AI-verktyget och dess tänkta användningen i systemet, vilket rapporterats i de två bilagda artiklarna. Den första artikeln visade att de meningsskapande processerna var ur fas mellan aktörerna, vilka växlade mellan sökande och oengagerade beteenden, och att användningen av gränsöverskridande objekt med rätt detaljnivå kan involvera oengagerade aktörer. Den andra artikeln identifierade beroenden mellan olika meningsskapande cykler, närliggande respektive avlägsen framtid, vilket belyser utmaningar med att koppla tillbakablickande insikter till framåtblickande föreställningar om systemet genom handling i nuet. Vid skapande av komplexa intelligenta system framstår mänsklig inblandning som oundviklig, och den andra artikeln belyser även hur AI kan förstärka mänsklig kognition och organisatoriska förmågor för att främja kreativ föreställningsförmåga kring möjliga och önskvärda scenarier av en avlägsen framtid.

Avhandlingen bidrar genom att bredda tidigare forskning om framtidsorienterat och kollektivt meningsskapande i utvecklingsprocessen av komplexa intelligenta system genom att presentera ett ramverk av närliggande och avlägset framåtblickande och intern och extern komplexitet. Det visar på nya insikter om hur kunskap flödar över systemnivåer och gränsöverskridande objekt kan användas under sådana utvecklingsprojekt. Sådana insikter kan vara praktiskt användbara för en välgrundad AI-introduktion i komplexa system och i samhället i stort. Dessutom bidrar den med en empirisk fallstudie av resan mot AI-introduktion i komplexa system till litteraturen inom innovationsledning.

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List of appended papers

Paper 1:

Särner, E., Yström, A., Lakemond, N., & Holmberg, G., (2023) *Prospective sensemaking in the FEI of AI-projects* [Accepted for publication in Research-Technology Management]
Presented at IPDMC conference, Lecco, 7-9 June 2023.
Under second round review in *Research-Technology Management*.

Authorship

Särner initiated the paper and was the principal author. Särner initiated and designed the study and took the main responsibility for the empirical work. The data analysis was done by Särner in discussions with the other co-authors. Theoretical framing was made together with the co-authors. Särner was responsible for paper submission to conferences. Särner was main writer in all parts of the manuscript.

Paper 2:

Särner, E., Yström, A., Lakemond, N., & Holmberg, G., (2023) *Utilizing AI in prospective sensemaking for desired futures: Outlining near- and distant-future sensemaking of complex system development* [Unpublished manuscript]
An earlier version was presented at the EGOS conference in Cagliari, 6-8 July 2023.
Extended abstract accepted to EGOS conference, Milan, 4-6 July 2024.
To be submitted to journal in spring 2024.

Authorship

Särner initiated the paper and was the principal author. Särner initiated and designed the study and took the main responsibility for the empirical work. The data analysis was done by Särner in discussions with the other co-authors. Särner developed the theoretical framing that was further elaborated jointly with the co-authors. Särner was responsible for paper submission to conferences and journal submission. Särner was main writer in all parts of the manuscript.

Table of content

1 INTRODUCTION 1

1.1 Background 1
1.2 Research purpose 4
1.3 Outline for kappa 5

2 THEORETICAL BACKGROUND 7

2.1 Complex systems 7
2.2 AI and innovation 9
2.3 Sensemaking 11

3 METHODOLOGY 15

3.1 Research approach and design 15
3.2 Data collection 17
3.2.1 Observations 17
3.2.2 Interviews 18
3.3 Documents 20
3.4 Data analysis 20
3.5 Quality criteria 21

4 THE EMPIRICAL CASE 23

5 PAPERS AND SUMMARIES 27

5.1 Paper 1 27
5.1.1 Paper outline 27
5.1.2 Contributions to thesis 28
5.2 Paper 2 29
5.2.1 Paper outline 29
5.2.2 Contribution to thesis 30

6 DISCUSSION 31

6.1 A tentative framework for prospective sensemaking in future complex intelligent systems 31
6.2 Near- and distant-future sensemaking 34
6.3 Internal and external complexity 35
6.4 Knowledge flows across system levels 36
6.5 Boundary objects 38

7 CONCLUSION AND FUTURE RESEARCH 39

7.1 Conclusion 39
7.2 Managerial implications 39
7.3 Limitations 40
7.4 Future research 40

REFERENCES 43

APPENDED PAPERS 49

Figures

Figure 1. System levels based on Trist (1981) 2
Figure 2 Founding theoretical fields for this thesis. 7
Figure 3. Important concepts within the grounding fields for this thesis. 14
Figure 4. Research process 17
Figure 5. Illustration of the actors in the social complex system 24
Figure 6. Illustration of misalignment in the collective sensemaking process in vignette 1 and vignette 2. 28
Figure 7. Interplay between near-future and distant-future sensemaking loops 30
Figure 8. Explicating types of prospective sensemaking in relation to complexity characteristics 32
Figure 9. Knowledge flow depending on prospective sensemaking and complexity characteristics 37

Tables

Table 1. The methods approach in the two papers. 15
Table 2. Observational data sources and use 18
Table 3. Interview data sources and use 19
Table 4. Document data sources and use 20
Table 5. Case information 25

1 Introduction

1.1 Background

Modern society relies on complex systems. Infrastructures such as electricity systems, transport systems, healthcare systems and communication systems which have been developed over centuries to provide societal services indispensable to everyday life. With time these systems have grown and become increasingly interconnected. Changes in society call for continuous adaptations of systems and new technologies are integrated into the existing systems, changing, and sometimes replacing parts of, or entire systems. This emphasizes the necessity to prepare the system for future needs and possibilities while building on the present state, incorporating decades of historical heritage.

This thesis focuses on the transition towards introduction of Artificial Intelligence (AI) in complex systems following the overall trend of AI integration into all parts of society, that with time can transform complex systems into complex *intelligent* systems (Lakemond & Holmberg, 2022). The empirical context for this thesis is the introduction of AI in energy systems. More specifically the development of an AI-tool for the planning of energy systems within the urban planning process is studied, including district heating/cooling and electricity. This is an example of a complex system, as described by Hobday (1998), characterized to be high cost, engineering intensive products and systems, with the need of a wide range of competences and knowledge (Hobday, 1998). Energy systems have characterizable long lifecycles where mistakes made in the design phase can become very costly to correct and thus linger in the system. The nature of complex systems adds uncertainty and risk to the innovation process and development often proceeds with incomplete information and emerging properties (Perrow, 1999). In fact, decisions in the planning of energy system are often made on very limited information, e.g. dimensioning of infrastructure cables that will serve future city districts for decades.

A definition of *complexity* in complex systems is that it cannot be fully comprehended by one individual expert nor communicated to another expert (Acha et al., 2004). To further clarify the complexity in complex system development, it can be helpful to distinguish between internal complexity and external complexity (Rousseau, 2019). Internal complexity specifically considers the technical complexity of the system with the corresponding organizational challenges, and external complexity regards contextual dependencies such as societal changes and dependencies on other systems. The development of energy systems depend on a fruitful collaboration between persons with different expertise, roles and perspectives as no one has the full understanding (Davies & Hobday, 2005). In this thesis, the studied AI-tool for planning of energy systems is developed in an inter-organizational collaborative project, which is typical

for development of complex systems (Hobday, 1998), integrating and making use of the new technologies and knowledge (Acha et al., 2004).

The energy system provides vital societal services and interruptions have big impact, implying that the system needs to be developed while maintaining operability and service. The energy system is in turn embedded in the social system of the city (Jones, 2014) influenced by local and national political decisions, regulations and public opinions. It is also interconnected with other systems and functions in the city such as transportation, water, and sewer or even e.g. school planning. Trist (1981) first presented three sociotechnical system levels to describe such systems (Saxena & McDonagh, 2022); macrosocial system, organizational system and work system. Work system is where the activities are performed and typically intend a single face-to-face group supported by management and specialists (Trist, 1981). The organizational system can be an entire cooperation or a public agency, while macrosocial system include communities, industrial sectors and institutions operating on an overall domain level (ibid). Figure 1 illustrates the system levels relevant for the empirical context in this thesis.

Macrosocial system	Social system – urban planning process
Organizational system	Complex system – energy system
Work system	Project – AI-tool development process

Figure 1. System levels based on Trist (1981)

AI is one of the most recent technologies to influence these systems, but despite its fast development, few organizations have to this point introduced AI-techniques in larger systems, with some notable exceptions like Google and Microsoft (Lewis et al., 2019; Ozkaya, 2020). AI concerns thought processes and reasoning as well as behavior (Russell & Norvig, 2016). When laid out on two axes, Russell and Norvig (2016) outline four categories of AI; thinking humanly, thinking rationally, acting humanly, and acting rationally. This thesis is concerned with AI within thinking rationally to give decision support to humans. This entails challenges to capture tacit knowledge, that can be difficult to express or knowledge that is not completely certain. To some degree, in this thesis the aim for AI is to act rationally as it, in the future, could recommend preferred configurations, and automatically discard others. However, in a complex environment it is unlikely that a single configuration will be universally the best solution, rather configurations will have different pros and cons.

The introduction of AI continues a more general trend of digitalization in society (Lakemond et al., 2024) and can be a way to grasp or handle the system complexity that goes beyond human capabilities. AI as a technology is often associated with both positive and negative expectations. Some are convinced that AI can contribute to

resolving some of the societal challenges that these systems are facing including urbanization and climate change, by predicting future developments and reduce some of the uncertainties and unknowns (e.g. Nishant et al., 2020; Vinuesa et al., 2020). Others raise concerns around e.g., loss of control and built-in biases (Dignum, 2019). This signals that further clarification and discussions around the role of AI and how to manage AI is important for a purposeful introduction of AI in these systems (ibid). AI's entrance in complex systems could also increase the uncertainties in the development phase (Lakemond et al., 2024). This can be connected to expectations and understanding of AI and the changes that the introduction of new technologies often bring to the development process including new roles and tasks (Berente et al., 2021). New actors with competence around AI enter existing industries that already have their established work traditions (Ozkaya, 2020). In other words, both knowledge in AI and domain knowledge are important for successfully introducing AI in complex systems (Piorkowski et al., 2021).

AI introduction can support management of uncertainties and unknowns in complex systems and can increase efficiency e.g. augment human cognition by analyzing big amounts of data (Jarrahi, 2018). However, AI can also introduce new uncertainties in the development including black box behaviors (Benbya et al., 2021; Berente et al., 2021; Ozkaya, 2020; Sagodi et al., 2023). This adds to the existing characteristics of complex systems with emerging behaviors and difficulty to maintain a holistic comprehension of the system. Still, the introduction of AI into complex systems could also reduce uncertainties with its analytic capabilities and reveal unknown behaviors and uncertainties (Jarrahi, 2018; Walker et al., 2003). Developers and operators in complex systems are familiar with management of uncertainty and emerging behaviors, and sensemaking is a well-known concept in complex systems literature to understand these characteristics of the system (Weick & Sutcliffe, 2015). However, the literature has mostly focused on the operations of existing systems and has thus mainly regarded retrospective sensemaking of analyzing and reacting to events that have happened in the past (Maitlis & Christianson, 2014). The transition towards complex intelligent systems implies a prospective perspective to anticipate future requirements and use of the system (Gioia & Chittipeddi, 1991; Jacobs et al., 2013) that has not been considered explicitly in this stream of literature. Prospective sensemaking can be seen as a process of making sense of the past to influence the future, taking place in the present. It links retrospective understandings to imaginaries of the future with enactment in the present (Kaplan & Orlikowski, 2013).

While no single actor can have overview of a complex system, sensemaking as a social and collective process is vital in system development. Hence, prospective collective sensemaking becomes central to understand the implications that the introduction of AI can bring in the context of developing future complex systems. Prospective collective sensemaking includes the interaction between actors to collectively develop

an understanding of the endeavor to be able to contribute to its realization (Gattringer et al., 2021; Stigliani & Ravasi, 2012). The early phases of an AI-project, i.e. the Front End of Innovation (FEI) is especially exposed to collective prospective sensemaking around AI. In this phase both how the specific AI technology works and is developed and its possible applications in the context of the specific complex system need to be understood. The sensemaking process entails iterations between collective and individual sensemaking, tacit and explicit knowledge, sensemaking and sense giving. These iterations and how actors take part in them collectively seem central for a purposeful introduction of AI in complex system development. It is important to acknowledge the point of view of stakeholders and actors situated in different part in and around the complex system for the AI introduction to contribute towards a desired future and avoid emerging behaviors (Gümüsay & Reinecke, 2022).

1.2 Research purpose

The motivations for the research outlined above can be summarized in four points:

- Enhance the understanding of the evolving phenomena of AI-introduction in complex systems by further exploration (Benbya et al., 2021; Berente et al., 2021; Von Krogh, 2018).
- Complex systems' vital role for modern society and the rapid changes it faces indicate the importance to further explore prospective sensemaking in addition to retrospective sensemaking for complex system development (Jacobs et al., 2013; Maitlis & Christianson, 2014).
- The introduction of AI into more traditional sectors provoke new collaborations, pointing at the importance to further explore the collective endeavor of integrating AI in existing systems (Gattringer et al., 2021; Lewis et al., 2019; Ozkaya, 2020; Stigliani & Ravasi, 2012).
- The exploration of how AI may use historical data that might not be representative or relevant for future scenarios, influence prospective sensemaking, and the transition to complex intelligent systems (Goto, 2022).

In sum, this thesis further examines the introduction of AI in complex systems and its implications on the development process. The aim is to *explore collective prospective sensemaking processes in the development of future complex intelligent systems*. In doing so, this thesis draws on two papers addressing the following research questions:

> *RQ1. How are prospective collective sensemaking processes enacted in the FEI of an AI-project?"*
>
> *RQ2. What role can AI-solutions play for prospective sensemaking in the development of complex system development?*

With this research aim, the thesis examines dimensions of prospective collective sensemaking in relation to complexities in complex intelligent system development and presents a tentative future-complexity framework. The empirical material that this thesis is based on mainly focuses on the sensemaking processes within the development of the AI-tool, i.e. the project level. Though the level of analysis is focused on the project level in the data collection, the need to also make sense of the energy system and the urban planning process has been present throughout the project, and thus also relating to AI-introduction on the complex system level and social system level (see Figure 1).

1.3 Outline for kappa

This thesis is a compilation thesis with two appended papers. In chapter 2 the theoretical background to the concepts founding the research is presented. In chapter 3 methodological consideration and the empirical methods that have been used are presented, covering both the appended papers and the thesis. This is followed by an extended description of the empirical case in chapter 4. Chapter 5 summarizes the findings from the two appended papers which is followed by a discussion that relates the findings to the thesis aim by presenting a framework in chapter 6. In chapter 7 conclusions as well as limitations, managerial implication and possible future research paths are presented.

2 Theoretical background

This thesis builds on three interconnected literature fields, see Figure 2. The complex product and systems (CoPS) literature is where the thesis has its main grounding, AI introduction is considered the focal point in the transition from complex systems to complex intelligent systems, and prospective sensemaking constitutes a theoretical lens to explore the development of complex intelligent systems.

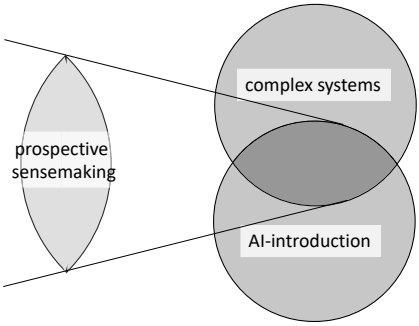


Figure 2 Founding theoretical fields for this thesis.

2.1 Complex systems

Complexity is a term present in many different research fields. Moreover, complex systems refer to a range of different kinds of systems. I therefore consider it important to establish what complexity and complex systems mean in this thesis. However, I use related fields and the more general definitions of complexity and complex systems to enhance and pin-down the understanding used in this thesis.

Simon (1962) described complex systems as a whole that is more than the sum of its parts. By combining elements on one level, they become building blocks of emerging properties on a higher level (Holland, 2014). Simon also distinguishes between the world sensed and the world acted upon, that he connects to problem solving described as a continuous translation between state and process description of the same complex reality. With this iterative process the system can be adapted purposefully. The complexity referred to in complex system literature can be regarded as organized complexity (Weaver, 1948). Organized complexity can be described with simplistic rules, but the vast number of variables soon makes it incomprehensible. This is different from unorganized complexity of e.g. the movement of gas particles, that mostly can be understood through statistics.

In the CoPS literature, complex systems are defined as the business of developing high-tech, high value business-to-business goods (Davies and Hobday, 2005). They are often produced in small batches and are highly customized and involves advanced engineering and software development (ibid). CoPS can be considered at the

organizational system level in Trist's (1981) classification (Figure 1). An important characteristic of complex systems is that changes in one part of the system can produce emergent behaviors and consequences in other parts of the system that are hard or impossible to foresee (Mayntz & Hughes, 2019; Senge, 1997). Emergence can be understood as the "action of the whole is more than the sum of the actions of the parts" and "selected aggregates at one level become 'building blocks' for emerging properties at a higher level" (Holland, 2014, p. 4). Moreover, the environment of CoPS is complex, i.e. Trist's (1981) macrosocial system level, with many involved actors and stakeholders (Hobday & Rush, 1999).

Complexity can be divided into internal and external complexity (Rousseau, 2019). Internal complexity is referred to as 'hard to describe but can have simple explanations' and often concerns technical complexity and the associated internal organizational challenges. The internal complexity frequently benefits from reductionist methods, where the system is divided into sufficiently manageable components that easily can be handled analytically (Beckerman, 2000). External complexity on the other hand is typically 'easy to describe but have complex explanations', like system thinking where the aim is to understand the overall dependencies (Senge, 1997). It often concerns dependencies towards the environment, stakeholder input, neighboring systems, or societal changes, and thus benefit from holistic methods for the management of such aspects.

Depending on the type of complexity in focus in the development of complex system, the emphasis needs to be on either reductionist or holistic methods. The growing complexity leads to a need for dealing with both internal and external complexity simultaneously and not only in different phases of the development. This implies an increased need to balance reductionist and holistic methods. Trying to understand the inner workings as well as regarding it a black box can together contribute to the understanding of complex systems by those developing and interacting with the system (Mack & Khare, 2016). While the inner workings, reactions, and behaviors of the system can never be understood fully, it is helpful to develop mental models of the system to ask the right questions and develop a feeling for the non-linear structures and dependencies. Observing the system from the outside will reveal emergent behaviors and although it might not be possible to deduce the inner workings causing the behavior, it can be useful for understanding boundaries and directions of the system's development.

The digitalization in complex systems is already advanced and has changed the scene in many ways. The complexity is increasing in systems that were already beyond human cognition (Davies & Hobday, 2005, p. 2; Rousseau, 2019). The digitalization has however also enabled development in the management of complex systems by improving the possibilities to monitor the systems, detect failures and simulate and

predict behaviors of the system ex-ante. These features were hard to predict a priori (Perrow, 1999) but have been important in making complex systems safer in combination with organization and development methods. AI-introduction can be seen as an ongoing continuation of this digitalization trend transforming complex systems into complex intelligent systems (CoIS) (Lakemond & Holmberg, 2022).

The methods applied in complex systems to cope with complexity beyond human cognition have been developed over time and can be useful also for handling characteristics of complex intelligent systems. One such insight is that when system complexity is beyond human cognition, everyone involved in the development organization need to be aware that they are a brick in a greater system. It is thus important to engage in system thinking, take a step back and consider the whole system, (Senge, 1997) and to visualize and raise awareness about interconnections and dependencies to prevent "not-my-task" behavior (Weick & Sutcliffe, 2015).

Another aspect is the importance of mental models (Senge, 1997) or cognitive maps (Weick & Sutcliffe, 2015). The mental models need to be updated and kept á jour with the present situations. As the complexity of the system is beyond a single person's comprehension, a culture that welcomes questioning and examination of our mental models is important to identify flaws and discrepancies (Senge, 1997; Weick & Sutcliffe, 2015). For this purpose, dialogue is well-suited where emphasis is on understanding each other rather than discussions where opposing views are presented and defended to find the best one. Assumptions should also be suspended to stay open for other points of views (Senge, 1997). This is important to create a continuous process of learning and knowledge building and not to simplify our understanding (Senge, 1997; Weick & Sutcliffe, 2015). Action to enhance the understanding and get more knowledge of complex situations before decision-making can also be emphasized (ibid). Many minds need to work together (Senge, 1997) and build a wider collective knowledge base (Weick & Sutcliffe, 2015). The role and function of mental models can be affected by some of the characteristics of AI. The inner workings of AI are for example hard to understand for many actors and can cause emerging behaviors that can be difficult to retrace. On the other hand AI can augment human cognition and thus support the formation of mental models (Jarrahi, 2018).

2.2 AI and innovation

AI is a rather vague concept without a universal definition and includes many different technologies and applications in many different fields. Machine learning (ML) is the most prominent of the AI technologies in recent years, driving much of the development in the field and it is the AI technology in focus in the empirical case of this thesis. ML is a bottom-up approach that learns from big amounts of data and its inner workings are often hard to understand for humans and described as a "black box" (Weber et al., 2023). With focus on AI that is 'thinking rationally', the difficulties with

ML becomes to build reasonable models of the world that are capable of incorporating tacit and uncertain knowledge into models (Dignum, 2019; Russell & Norvig, 2016). Walker et al. (2003) define uncertainty as the *"deviation from the unachievable ideal of completely deterministic knowledge of the relevant system"*. New knowledge can both increase and decrease uncertainty and previously unknown or overlooked uncertainties can be revealed when new knowledge of the system is mastered (ibid). Dimensions of uncertainty that Walker et al. (2003) describe include the nature of uncertainty as caused either by the imperfection of our knowledge, epistemic uncertainty, or by the inherit variability of the phenomenon described, variability uncertainty. In AI introduction in complex systems both epistemic and variability uncertainty is prevalent. The epistemic uncertainty can be seen as situated as it varies significantly between the different actors involved. The variability uncertainty stem mainly from social and economic dynamics on a macro-level and technological surprises.

The perceived increase of uncertainty in the development of AI can be associated with the fact that it is an evolving phenomena making it difficult to foresee exactly what it implies for the organization and the operations it is introduced in (Berente et al., 2021). Moreover, AI often has dynamic, tentative and incomplete functionality causing high volatility in the development and use (Ångström et al., 2023), it is hard to specify the functionality up-front and it has the ability to learn and act autonomously (Berente et al., 2021; Ozkaya, 2020). The software development process needs to find ways to handle new characteristics of AI such as data centricity, explainability, verifiability and change propagation as it has tackled challenges in the past with e.g. usability and data security. Some aspects of AI might call for changes and updates of current development methods (ibid). AI-enabled systems can be hard to specify up-front which is true also for many traditional software systems where uncertainty and "unknown unknowns" are problematic. However, it becomes more prominent when AI is introduced into a system as the requirement specification and the problem specifications often are the same. Another aspect is that dependencies are increasingly difficult to trace, especially data dependencies, that increases the need for understanding and management of how changes propagate and emerging behaviors develop in the systems. ML components often tend to be more entangled than traditional software modules which makes it harder to maintain modular boundaries (Ozkaya, 2020). Reduction of uncertainty and error localizations when ML is introduced in existing systems is often harder than anticipated (Amershi et al., 2019).

ML introduces new tasks and roles in the development process (Amershi et al., 2019) and new actors that need to make sense of AI and the development process which adds to the uncertainty (Benbya et al., 2021; Ozkaya, 2020; Sagodi et al., 2023). Roles involved in the development of ML-systems include data scientist, software engineers and operational staff. The data scientists build and train models and test them. The

software engineers integrate the trained model into a larger system usually relying on agile work processes that need to be adapted to the new challenges that the AI introduction brings (Ozkaya, 2020). Challenges are related to e.g. discovering data, managing, and handling versions of the data and adapting and standardizing working processes. The operational staff deploy, operate, and monitor the system. The different roles have different perspectives and use different languages which can be a cause of miscommunication (Lewis et al., 2019; Piorkowski et al., 2021).

Beyond the challenges within software development, research has also demonstrated challenges in the wider context of the developing organization when AI is introduced. In the FEI of complex system development many actors and stakeholders are introduced to novel collaborations, e.g. with AI developers. The diversity of the actors increases when considering the development process beyond software development, with increased variation in knowledge and expectations of e.g., AI functionality. For example, AI developers have been shown to disregard domain knowledge (Sambasivan & Veeraraghavan, 2022) and the actors need to bridge gaps in knowledge to overcome uncertainties, build trust, and align expectations with their communication (Piorkowski et al., 2021). ML has been shown difficult to explain to non-technical actors in a way that they can relate possibilities with AI to challenges in their own field (Ayobi et al., 2021). The FEI-phase that is already noted for high uncertainties faces additional challenges within the development team as AI is introduced.

2.3 Sensemaking

Sensemaking as a method to understand complexity has been elaborated in complex systems literature with focus mainly on operations (Weick & Sutcliffe, 2015). However, the introduction of AI triggers a need for a new perspective on how to make sense of the systems prospectively as they transform into complex intelligent systems. The introduction of AI can be anticipated to show similarities to other technology shifts (Orlikowski & Gash, 1994), but the characteristics of AI also suggest that sensemaking processes can be influenced e.g., AI connects historical data with predictions in a new and more distinct way. It also explicates explainabilty issues where actors not directly involved in the development of AI need to understand what is going on.

Sensemaking has traditionally been considered a retrospective process to give meaning to lived experiences (Gattringer et al., 2021; Weick, 1995). It is a social and ongoing activity (Maitlis, 2005) in which enactment is central to influence the structures, constraints, and opportunities that in turn can progress the sensemaking process (Weick, 1988). It is a process of social construction when individuals interpret and explain cues in their environment, creating ‘accounts’ that allow dealing with uncertainty and ambiguity (Maitlis, 2005). Sensemaking can be described as an interactive process involving iterations between individual and collective sensemaking (Stigliani & Ravasi, 2012), between tacit and explicit knowledge (Dougherty et al.,

2000), and sensemaking and sensegiving (Medlin & Törnroos, 2014). While complex systems development needs to be grounded in retrospective knowledge due to existing design based on historical decisions, the development process also entails prospective sensemaking in imagining future use and possibilities (Jacobs et al., 2013). This is a concept that more recently has been given increased attention (Maitlis & Christianson, 2014) and can be described as *"the conscious and intentional consideration of the probable future impact of certain actions, and especially non-actions, on the meaning construction processes of themselves and others"* (Gioia et al., 1994, p. 378). Complex systems development can thus be seen as the process of connecting past events with future expectations and desires by action in the present. Prospective sensemaking relate to imagination of desired futures which can be different in nature depending on the distance in time (Gümüsay & Reinecke, 2022). Near-futures can often be predicted more easily while distant futures are harder and might be less influenced by the past and present (Rindova & Martins, 2022).

Organizational sensemaking is a social and collective process which is prominent in complex system development where no one has the complete picture. Collective sensemaking in a development team can be described to start with the *exposure* of ideas, understandings and assumptions where individual experiences and knowledge are shared in the group (Stigliani & Ravasi, 2012). This is followed by *articulation*, where experiences expressed and embodied in artifacts are collectively ordered for emerging new understanding, providing access to each other's experiences and *elaboration*, where provisional understandings are tentatively integrated into more complex interpretations through conversations often with support of the embodied artifacts. Articulation and elaboration within a development team can be contrasted with practices such as *co-opting* identified in collaborations between functions to convince other groups by deliberately using concepts and language of that group and *repurposing* by utilising the practices of the other group in hope that they will reach the same conclusions within their own convictions (Beverland et al., 2016). Gattringer et al. (2021) identified the *articulating phase* to be especially important in cross-industry settings to form understanding of the individual experiences exposed in the previous phase, as it entails more unfamiliar information to many actors. This is also the phase where facilitators were identified to have the most impact for a successful prospective sensemaking by managing the information flow and impeding dominating actors to impose their view in the group that could inhibit the prospective sensemaking process. *Sensegiving* can be seen as the last phase where a development team try to influence sensemaking process of external stakeholders (Gioia & Chittipeddi, 1991; Stigliani & Ravasi, 2012). Sensegiving can also be used by stakeholders and leaders in the process (Maitlis & Lawrence, 2007) and while stakeholders seem to engage in sensegiving when they perceive it to be important and sense a need to support others in making sense, leaders often engage in sensegiving when the multiplicity of involved stakeholders increases the ambiguity in the situation.

The sensemaking process has mostly been described to rely on narrative practices, however the role of boundary objects has more recently been introduced as complementary in the sensemaking process (Carlile, 2002). Boundary objects are artifacts used to communicate across social boundaries, between social worlds (Star & Griesemer, 2015) and is one of many AI concepts that made its way into organizational theory (Csaszar & Steinberger, 2022). The related concept of material memory refers to artifacts that offload the human mind where ideas and understandings can be dispatched and returned to later bridging gaps in time and knowledge, as well as between individuals (Stigliani & Ravasi, 2012), relating to distributed cognition where human cognition is offloaded by artifacts (Hutchins, 1995; Secchi, 2010) and augmenting human cognition (Raisch & Krakowski, 2021). In ML-development analogical narratives have been shown to serve as boundary objects to explain the technology to actors with different backgrounds (Ayobi et al., 2021). ML-applications with its ability to find patterns and to process large amounts of data can also serve as tool to rely on in sensemaking (Secchi, 2010). It can work as a material memory, as a support for human cognition, and as a boundary objects in collective sensemaking and thereby augment human cognition (Raisch & Krakowski, 2021). Boundary objects embody work and knowledge and can be used for problem solving (Bechky, 2003). Caccamo (2020) presents four affordances of boundary objects important for collaborative innovation; convergence, generativity, socialization and collaborative learning. Convergence of separate experiences enable deep knowledge sharing and cooperation. To support the generative nature of innovation, boundary objects need to allow for open-ended modifications. Socialization refers to the creation of shared practices, cultures and values by selection and modification of objects. Collaborative learning refers to new skills and self-awareness developed when interacting with shared objects and reflexivity triggered by perspective making and perspective taking.

This thesis builds on the three theoretical fields described above and contributes to the intersecting area in the diagram in Figure 3 where the findings draw on important concepts from theories within all three fields.

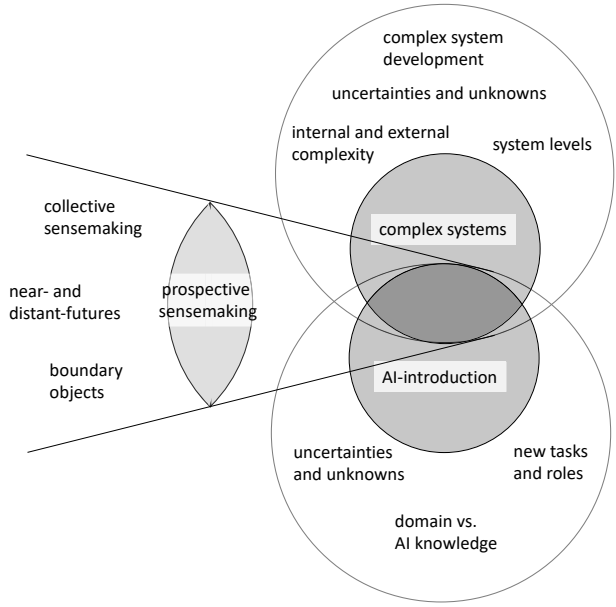


Figure 3. Important concepts within the grounding fields for this thesis.

3 Methodology

This thesis builds on a single case study reported in the two appended papers. The data collection and data analysis from the case study is the foundation for both papers. However, different data sources have been more important for one or the other paper and different data analysis approaches have to some extent been applied to answer the research questions as outlined in Table 1.

Table 1. The methods approach in the two papers.

	Paper 1	**Paper 2**
Title	Managing asynchrony in prospective collective sensemaking in AI-driven innovation	Utilizing AI in prospective sensemaking for desired futures: Outlining near- and distant-future sensemaking of complex system development
Research question	How are prospective collective sensemaking processes enacted in the FEI of an AI-project?	What role can AI-solutions play for prospective sensemaking in the development of complex system development?
Data collection methods	Interviews Observations Documents	Interviews Observations Documents
Analysis methods	First and second order coding Narrative analysis using vignettes	First and second order coding Narrative analysis
Level of analysis	Project level	Project level Complex system level Social system level

3.1 Research approach and design

The study of a new phenomenon like AI-introduction in complex systems needs to be grounded in real-world observations that can bring new insights, advance our current theoretical understanding and give practical guidance in real-world situations (Fisher et al., 2021). AI-introduction is a general ongoing trend in society that we all need to relate to. The integration of AI in sectors previously unaffected becomes an especially relevant concern for new actors that then need to relate to it. I thus found the individual understanding of AI a highly interesting component to examine and how it influences the collaboration between different actors in their work with AI-introduction in complex systems development. By exploring "how people describe things and experience them through their own senses" (Patton, 2015, in Merriam & Tisdell, 2016, p. 9), this study takes a constructivist approach acknowledging multiple experiences of reality. The chosen theoretical lens of sensemaking aligns with this approach as it puts

focus on the individual construction of frames that influence action that is inherently differing between person, situation and context.

To explore this new phenomenon of AI-introduction in complex systems that few researchers have studied previously, an explorative case study was found appropriate. Qualitative case studies are suitable when it comes to studying complex phenomena in contemporary contexts (Baxter & Jack, 2008). The case of Mälarporten was purposefully selected for its context with the involvement of many actors with fundamentally different backgrounds. This provided high likelihood to reveal valuable insights regarding the research purpose and research questions (Eisenhardt, 2021; Merriam & Tisdell, 2016). Energy systems and other utility systems are classically regarded as complex systems and thus display many of the associated characteristics (Acha et al., 2004). Their embeddedness in society and the current challenges related to societal and technical changes, such as urbanization, electrification of cars, and expansion of renewable energy sources further motivated the choice as a relevant context to study.

The forward-looking approach to anticipate future needs in such a future-oriented project also reflects a prescience approach. Corley and Gioia (2011) define prescience as the process of *"discerning or anticipating what we need to know and, equally important, of influencing the intellectual framing and dialogue about what we need to know"*. The case seemed apt to contribute to anticipation of future needs in management and design of complex systems when AI is introduced (Stilgoe et al., 2013). The data collection in the case study has been performed at the project level, and the level of analysis has been mainly at the project level i.e. the AI-tool development project. However, interrelating with the complex system level, i.e. the energy system and occasionally also social system level is discussed i.e. the urban planning process (Yin, 2018). The system levels are illustrated in introduction (Figure 1).

The study had a strong inductive approach in the beginning to let the empirical material guide the direction of the study rather than confirming a given theory (Merriam & Tisdell, 2016, p. 17). However, the theoretical boundaries of the research group have guided the study, and the research group has its starting point in complex systems, innovation, and organizational studies. With time, more abductive reasoning was applied to relate the empirical data to existing theories of sensemaking (Sætre & Van de Ven, 2021). The study provided rich descriptions and in-depth understanding around the context and conditions of AI introduction in complex systems. The objective is to utilize case insights to contribute to further development of theory.

3.2 Data collection

The core data sources in the case study have been observations and interviews. The findings from interviews and observations were triangulated and complemented with documentation about the case and the sector. A timeline of the data collection in relation to the case project and the writing process of the two papers can be seen in Figure 4.

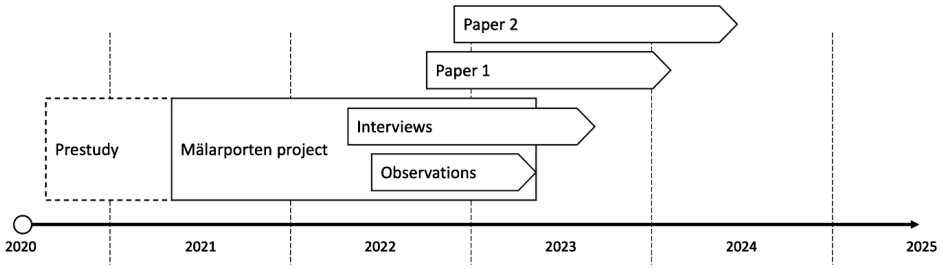


Figure 4. Research process

The initial contact with the project was through the municipality of Västerås were one of the participants of the project became a key informant granting access to the other participants and to the project meetings. Two initial interviews with the key informant were held together with my supervisors. The following 24 interviews were conducted by me alone as well as the observations in 24 meetings, including sprint meetings, work meetings and meetings with external stakeholders. Due to the pandemic and the geographical distribution of team members, all interviews except one was held online. All project meetings were likewise organized online, i.e. all meeting participants connected individually to a digital meeting notice. The initial data analysis was made with support of my supervisors. As outlined in more detail in 3.4 Data analysis, I performed 1st and 2nd order coding of the interviews that was later iterated in the writing process of the papers (Miles et al., 2020).

3.2.1 Observations

I performed non-participant observations to gain general understanding of project, the situations and context of the case (Ciesielska & Jemielniak, 2018, p. 40). This gave me an understanding of the sensemaking process and potential topics to explore as the project progressed and I could pick up incidents not mentioned in the interviews. Impressions from observations can help revise the research problem that in turn requires different methods and theoretical framings making it appropriate for this explorative study (Ciesielska & Jemielniak, 2018, p. 34). The meetings I observed were sprint planning meetings, sprint follow up meetings, sprint demo meetings where mostly the core project team was present. Sometimes I also had the possibility to do observations at work meetings with some of the participants. Moreover, I did

observations at reference meetings where external stakeholders from e.g. universities and other municipalities were invited. The online format suited the non-participant observation approach of not engaging in the meetings and minimized the observer influence. During the meetings, I took extensive notes trying to describe what happened rather than placing judgement (Ciesielska & Jemielniak, 2018, p. 41). After the meetings, I noted down my reflections and attached to the descriptive notes. See Table 2 for more information. The observation sessions also gave opportunity to identify further interview subjects for the study.

Table 2. Observational data sources and use

Data source	Type of data	Use in analysis
Observations (~30 min meetings, total ~5,5h)	*Notes from observations at biweekly team* meetings, where interactions and collective sensemaking process was followed over time.	Notice significant behaviors and occasions relevant to the sensemaking process. Triangulate information from interviews.
(~2h meetings, total 4h)	*Notes from observations at reference meetings*, where the results and experiences so far were presented and feedback and discussions from external parties with interest in the project.	Notice significant behaviors and occasions relevant to the sensemaking process. Triangulate information from interviews.
(1h-1,5h meetings, total ~5h)	*Notes from observation of work meetings*, where technical solutions or usefulness was discussed.	Notice significant behaviors and occasions relevant to the sensemaking process. Triangulate information from interviews.

3.2.2 Interviews

25 Interviews were held with the 14 of the participants in the project to get access to their thoughts around AI, the project and situations unfolding in the meetings that could then be compared and contrasted. This included the main participants in the project and some more peripheral actors. The first interviews held with a key informant in Mälarporten were explorative to understand if the case would fit our research interests. When the key informant had provided access to the core project participants, I interviewed all the main actors in the project to capture the different views on the AI development. Snowball sampling strategy also gave access to more peripheral participants that were less frequently involved in the project. I strived to cover as many roles and responsibilities as possible in the project and returned to the core project group during and after the project ending for follow-up interviews to get an

understanding of developments over time. I personally transcribed all interviews which gave a god familiarity with the empirical data. See Table 3 for more information.

The interviews were semi-structured to guide the conversation and understand the interviewees better and at the same time be able to adapt the interview to the person and situation (Merriam & Tisdell, 2016, pp. 110-111). Themes and general questions were prepared but not followed rigorously or in any specific order. In addition, situations noted in observations were brought up in the interviews to get a deeper understanding of the development situations occurring when AI is involved (ibid). The interviews included questions such as if they had experienced any surprising behaviors or how the communication has worked with different actors. At the start of the interviews the purpose of the study was explained to the interviewees providing for them asking clarifying questions and enabling them to contribute to the project more actively (Pawson, 1996). Moreover, consent for recording was sought at the beginning of each interview.

Table 3. Interview data sources and use

Data source	Type of data	Use in analysis
Interviews (12 interviews, ~12h)	*Initial interviews* with main actors (project management, project coordinator, developers, clients, and experts) to capture individual impressions, perspectives, knowledge, and backgrounds. Held during summer and early autumn 2022.	Familiarize with the project and the context.
(7 interviews, ~5h)	*Additional interviews*, to get clarifications and additional information held as follow up interviews and with additional actors in the project. Held during winter 2022-2023.	Triangulate information from other interviews and observations.
(6 interviews, ~3h)	*Concluding interviews*, conducted at the end of the project with the main actors to give further clarifications and follow up the initial impressions and perspectives of the actors. Held spring and summer 2023.	Triangulate information from other interviews and observations.

3.3 Documents

Documents around the empirical case were collected to deepen the understanding. Some internal project documents were also accessed that could contribute to further understanding of the case project. The secondary data was collected throughout the study, official documents such as project reports and official statements, and internal documents such as project descriptions, working documents and other documents produced in the project. See Table 4 for more information.

Table 4. Document data sources and use

Data source	Type of data	Use in analysis
Secondary material (5 documents)	*Internal documents*, e.g., visualizations of the tool and project documents.	Give insight in the sensemaking process over time. Support and triangulate information from interviews and observations.
(10 documents)	*Public documents*, e.g., online publications, pre-study report	Familiarize with the project and the context. Support and triangulate information from interviews and observations.

3.4 Data analysis

During the data collection a tentative interpretation of the material was made that sparked the ideas for both paper 1 and paper 2. In the process of writing the papers these interpretations were revised and scrutinized as the data analysis progressed and verifications or invalidations of the initial interpretation further advanced the analysis.

The interviews were transcribed and through thematic coding including first and second order codes with an effort to stay true to the material collected in the first order codes (Miles et al., 2020). Keeping the first order coding as close to the informant's choice of words as possible and comparing these in a second step to find similarities and differences and thereby grouping them to a more manageable number of themes. These were sometimes analyzed to find "aggregate dimensions" as the highest abstraction level (Gioia et al., 2013). Effort was then put into seeking connections with previous research to further elaborate and clarify the theoretical contribution (Van de Ven, 2007). To ensure the traceability all material collected has been saved in a systematic way building up a case study data base including preparatory material as interview guides, raw materials as interview recordings, transcripts, and documents as well as memory notes, early reflections and write-ups (Eisenhardt, 1989). In the

process of writing the papers the data analysis varied to some degree as summarized in Table 1.

In paper 1 the analysis consisted of constructing a more focused data structure (Gioia et al., 2013) which was supported by the creation of narrative vignettes of significant sensemaking events that helped pin-pointing the core issues in the rich empirical material (Barter & Renold, 1999; Miles et al., 2020). Paper 2 utilized the data from a process perspective (Langley, 1999), entailing a narrative approach, focusing on providing a thick description of the project. Two storylines were written that described the sensemaking of near- and distant-future. These were broken down to key events represented in an illustration that were perceived as critical to the sensemaking loops. This was also supported by 1st and 2nd order thematic coding (Miles et al., 2020). Then, interrelations between the loops were identified with support from the coding. The outcome was a narrative description and an illustration of near- and distant-future sensemaking loops.

3.5 Quality criteria

Each research approach demands tradeoffs between accuracy, generality, and simplicity (Weick, 1979). For upholding trustworthiness in qualitative research Lincoln and Guba (1985) present four criteria to relate to; credibility, confirmability, transferability and dependability. The main objective of this research is to provide analytical generalizability by providing in-depth insights from an illustrative case, providing ample opportunities for practitioners and academics to learn about sensemaking and AI introduction in complex systems and get inspired. Procedures and documentation of process from data to findings attempts to provide the possibility for the reader to judge the transferability of the findings to other contexts. The Mälarporten case was selected for its potential to demonstrate an important phenomenon and provide valuable insights to the research purpose. A case that provides rich material to learn and generalize from and a narrative description has attempted to give sufficient understanding of the case to the reader to permit comparison with other cases (Miles et al., 2020). The narrative and visual mapping data analysis strategies (Langley, 1999) supported staying relatively close to the data. Any additions to theory have been explicitly stated and related to prior theoretical understandings and limitations and further studies for confirmation of the findings have been suggested (Miles et al., 2020).

The use of interviews and observation as the main data collection methods have contributed to the credibility by providing a 'thick' description of the studied case (Miles et al., 2020). Triangulation by comparing and contrasting data from the different sources, supplemented also by the collections of related documentation has further supported credibility (Merriam & Tisdell, 2016, pp. 244-245). Thematic analysis with first and second order coding was applied (Kiger & Varpio, 2020) which

has ensured traceability from theoretical constructs to empirical data (Merriam & Tisdell, 2016, pp. 251-252).

Dependability has been strived for by a transparent approach towards the participants in the study, explaining the research background and aim in the beginning of the interviews (Miles et al., 2020), draft version of the papers were offered to the participants when showing interest for the results (Merriam & Tisdell, 2016, p. 246). This built trust with the participants in the study. Interviews with as many actors as possible involved in the project was strived for to collect data that would cover different perspectives of the project (Miles et al., 2020). Moreover, the main actors in the project were returned to adding to the dependability of the collected data.

The methodological choices have been described with care to aspire to conformability, strive for neutrality, and refraining from unacknowledged bias in the process of generalizing the findings and relating them to theory (Miles et al., 2020). The use of thematic analysis has demonstrated traceability from theoretical constructs to empirical data (Merriam & Tisdell, 2016, pp. 251-252). Reflexivity was also important throughout the data analysis trying to reveal personal assumptions and biases reflecting in the analysis and findings (Merriam & Tisdell, 2016, p. 249). This has been done through discussions with other researchers within the research group. Iterating interpretations of the data with the supervisors that are also co-authors on the papers and feedback from discussions with international experts. Presentations at research conferences have also resulted in discussions and feedback valuable to further develop the understanding to the empirical material. The process of how collecting and analyzing data has been described transparently to further ensure conformability (Miles et al., 2020).

4 The empirical case

The qualitative case study (Merriam & Tisdell, 2016) that this thesis draws from is a development project within an initiative called “Kraftsamling AI”. It is an initiative to enhance the knowledge of AI in the civil engineering sector in Sweden financed by the financing agencies Vinnova, Formas and the Swedish energy agency. Västerås stad together with MälarEnergi - the local and city owned energy company, and ABB - a leading company in electrification and automation, applied for funding to use Mälarporten as a case for utilizing AI in city planning and is one of three projects that were financed. The other projects focusing on AI in operations and AI in the construction site.

Mälarporten is a new city district in the center of Västerås, Sweden, replacing an industrial area with many small industries. Within the area the most prominent building is the central station and travel hub, which is managed as a separate project. The name Mälarporten, “the gate to Mälaren” refers to the connection of Västerås to the outside world, by trains and busses at the travel hub. It also refers to connecting the city center with the lake Mälaren and lakeside areas of the city that today are cut off by the railroad tracks and crossing the tracks when the station’s walkway is closed requires a significant detour. The project has an outspoken focus on sustainability and to include innovations in the design.

The civil engineering sector is a sector involving many different actors, from small family businesses to large construction firms, a mix of public and private actors, and with a high degree of regulation on how things should be done. In the development of such a complex system as the energy system, there are many regulations to adhere to. This brings challenges connected to decision making on different levels, unique project conditions, collaboration between diverse actors, etc. It is also a relatively traditional setting where you do things like you have always done and a lot of knowledge is held tacit among experienced actors. In the planning of new city districts many decisions need to be made around the infrastructure. However, in the early phases of planning very little information is available, leading to decisions around big investments being taken on the basis of very limited information. As for complex systems in general, every project in the civil engineering sector is unique and therefore it can be difficult to draw parallels and transfer learnings from earlier projects. Decisions regarding the design of the area are taken within the urban construction process involving many different actors from the region and the municipality, the city of Västerås, as illustrated in Figure 5. It is largely private or public landowners that decide what infrastructure they need, limiting MälarEnergi’s ability to influence how to design the system as they are required to provide what the customer asks for. This makes it hard for MälarEnergi to take a system’s perspective in the infrastructure planning where synergies between buildings can be exploited, e.g., one building can produce solar energy for others, one building can install a battery and store energy for many buildings, or electricity peak

when starting an escalator can be covered by the charging posts for electrical vehicles in the garage next door. The municipality has some possibilities to influence the outcome by how the detailed plan for the area is formulated and by placing requirements for building permits. However, they do not have expert knowledge of the energy system and they rely on MälarEnergi, the city owned energy company, for that.

The aim of the studied project was to provide decision support in the early phases of the urban construction process using AI technology. AI was seen as a possibility to increase the available information that today, as mentioned, is very limited. The project brought together actors from the city of Västerås, MälarEnergi, ABB, Utilifeed - a company providing a SaaS platform for smart control of district energy utilities, and RenBloc - a startup company providing calculations of the energy mix and CO_2 emission. It also included project managers connected to Smartbuilt Environment with acting project managers from Data Edge. The projects position with the social complex system is illustrated in Figure 5.

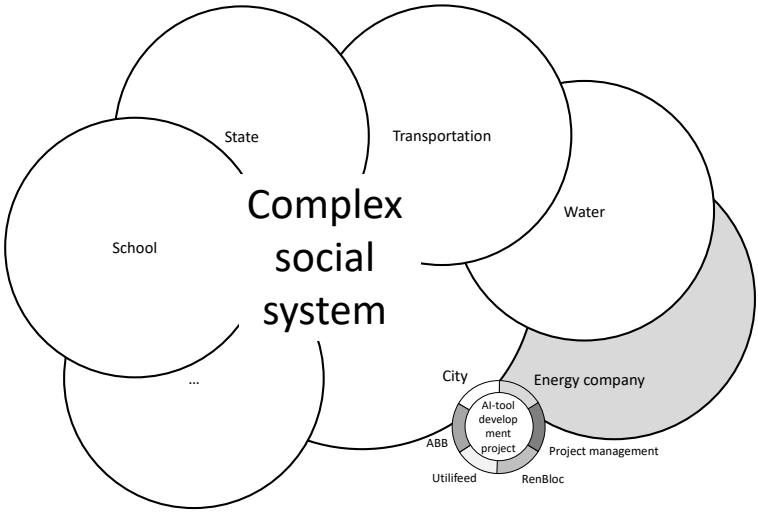


Figure 5. Illustration of the actors in the social complex system

In the beginning the aim of the project was very open-ended with the task to develop AI for the city planning process and the project team explored different alternatives before deciding to focus on an AI-tool for planning of energy systems. When this had been decided, the actors went on to collectively explore what the AI-tool should do and could look like to provide decision support, including CO_2 impact, for planning of new city districts. They also discussed issues related to how such a tool should or could be used, leading to discussions on who would be the best owner of the tool and discussions around changes to business models that would be needed to efficiently be able to use the tool.

The tool uses historical consumption data for electricity and district heating and cooling and combines it with historical energy production and weather data. The consumption and production data need to come from the same year as both are highly dependent on the weather. The consumption data is divided in categories for e.g., apartments, offices, schools, and business premises. It is then mapped for housing mixture of the new city district using the planned square meters of the different categories that are available in the city plan. The tool simulates different energy system configurations for the city district to predict CO_2 impact, peak loads etc. for each configuration that can be used as decision support in the urban planning process. The outcomes from the different configurations are then compared and modified to arrive at a preferred design. More detailed information on the case can be seen in Table 5.

Table 5. Case information

Case Mälarporten	
Project time span	April 2021- March 2023
Scope	Enhance knowledge of AI in the civil engineering sector. Develop a prototype for an AI-tool to be used in the planning of the energy system of Mälarporten, Västerås, Sweden.
Participating organizations	Västerås stad The energy company in Västerås, owned by the city. Acting as costumer/user of the tool and as property owner and builder. Providing knowledge on the city planning process and contributing with consumption data for public buildings like schools and offices. MälarEnergi Acting as costumer/user of the tool, project coordinator, and system integrator for ABB and Utilifeed's modules. Providing knowledge on the city planning process and consumption data for heating/cooling. ABB Provider of an optimization platform, control system and module of the electrical system Utilifeed Provider of a module of the heating/cooling system and developing their platform to the needs of the project. RenBloc Provider of a module for CO_2 emission integrated through ABB's module. Adapting it for the needs of the project.

Case Mälarporten	
	Smartbuilt environment Providing project management with acting project managers from Data Edge.
Design process	Online meetings The project started during the Covid-19 pandemic and the project meetings were organized online. Later a few physical meetings were held, the first in august 2022, but work continued to be organized online as the participants were distributed in different cities in Sweden. Team meetings The project followed an agile approach, with sprints of three months beginning with a sprint startup meeting and ending with a sprint demo. During the sprints biweekly follow-up meetings were held. External meetings Every 6 months a reference group meeting was organized online for external stakeholders and an additional external information meeting was organized mid-way into the project. The project was concluded with a final conference where the final result was presented to external stakeholders.

5 Papers and summaries

5.1 Paper 1

Prospective sensemaking in the FEI of AI-projects

Presented at IPDMC conference, Lecco, 7-9 June 2023.
Accepted for publication in Research-Technology Management.

5.1.1 Paper outline

Introduction of artificial intelligence (AI) in the development of new complex systems poses challenges for less technically oriented organizations and risks undesired prolongation of the Front End of Innovation (FEI) phase, due to e.g. diverging understandings of AI-functionality and requirements among involved actors, and the difficulties of determining the usefulness of AI in such a complex setting. This paper explores a cross-industry project developing a functional prototype of an AI-tool for planning (complex) energy systems in new city districts, engaging both system (domain) actors and AI-developers by examining the research question: *How are prospective collective sensemaking processes enacted in the FEI of an AI-project?*

The analysis of the prospective, collective sensemaking processes is based on first and second order coding of the interview data. The presentation of two vignettes from the project captures two critical sensemaking episodes also illustrated in Figure 6, outlining the different actors' sensemaking behaviors. Vignette 1 illustrates the discussions taking place in the project meetings in the first part of the project around ownership of the AI-tool. These engaged mainly the domain actors 'seeking' collective sensemaking to understand the future use of the AI-tool and influence on the home organization, *articulating* their understandings and trying to *elaborate* to agree and settle the discussion. Despite participating in the same meetings, the AI-developers were 'disengaged', finding the discussions not directly related to their work at hand but still participating with *sensegiving* for the domain actors' sensemaking process. Vignette 2 illustrates that later in the project when focus shifted more towards the technical development, the AI-developers moved from *articulating* to *elaborating*. However, the domain actors expressed frustration and confusion of not following the project progress while still engaged in *articulating*. Project management responded to this reaction by returning to low-fidelity prototypes illustrating the user interface and the data flow of the AI-tool 'supporting' the collective sensemaking and allowing them to *elaborate*. Soon after this, the first demonstrations of the functional prototype occurred. This further 'supported' domain actors' awareness of the project progress. Moreover, it supported the *elaboration* of the AI-developers. However, soon thereafter, the domain actors chose to leave the project as they felt they could not contribute and thus 'disengaging' collective sensemaking.

As indicated through the different movements illustrated in Figure 6, the two vignettes uncovered misalignment of sensemaking processes between system actors and AI-developers as they strived to reconcile multiple understandings of the future functionality of the AI-tool. Alternation between 'seeking' and 'disengaging' prospective sensemaking behaviour were observed during the project. The vignettes also highlight how prototypes can bridge AI-related knowledge discrepancies and support alignment of collective sensemaking processes and engage disengaged actors. Low fidelity prototypes helped domain actors to elaborate on what the AI-tool could do related in the challenges in their field as demonstrated in vignette 2. However, the level of detail could be an important factor to also engage the AI-developer in the collective sensemaking in this phase as they were reluctant to do so. They however sought collective sensemaking as the functional prototype was further developed and possible to demonstrate also demonstrated in vignette 2.

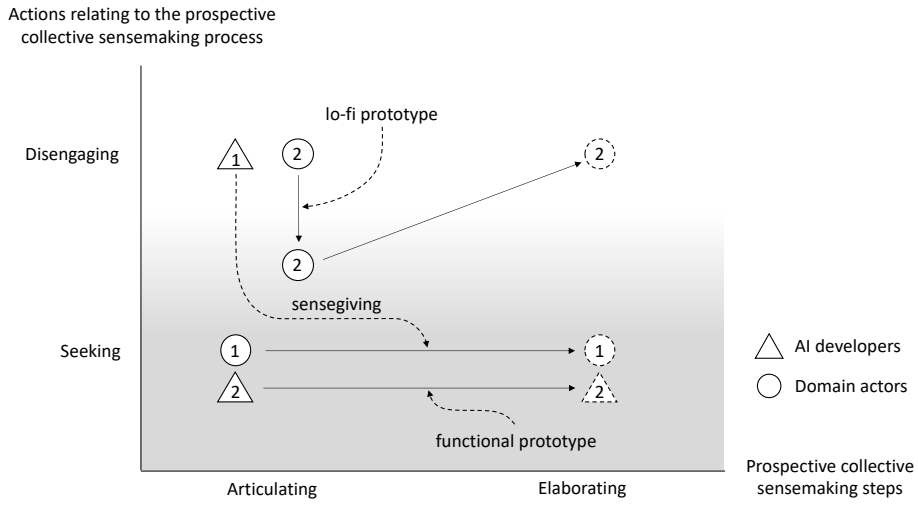


Figure 6. Illustration of misalignment in the collective sensemaking process in vignette 1 and vignette 2.

5.1.2 Contributions to thesis

This paper contributes to the thesis by adding insights on the research question "*How are prospective collective sensemaking processes enacted in the FEI of an AI-project?*". It contributes with insights mainly on the project level of analysis. The identification of the approaches: seeking, supporting, and disengaging towards collective sensemaking informs our understanding of the collective prospective sensemaking process in the development of future complex intelligent systems. The identification of boundary objects of different levels of detail and their potential support of the collective sensemaking process between actors with different knowledge backgrounds. This also enhances our understanding of the collective prospective sensemaking process by relating boundary objects to the different sensemaking approaches identified.

5.2 Paper 2

Utilizing AI in prospective sensemaking for desired futures:
Outlining near- and distant-future sensemaking of complex system development

An earlier version was presented at the EGOS conference in Cagliari, 6-8 July 2023.
Extended abstract accepted to EGOS conference, Milan, 4-6 July 2024.
To be submitted to journal in spring 2024.

5.2.1 Paper outline

This paper examines the role of AI-applications in the prospective sensemaking process for development of complex systems, including the relation between historical data of the past and imagined and desired futures. The research question posed is: *What role can AI-solutions play for prospective sensemaking in the development of complex system?*.

The findings distinguish between two separate, but interrelated, sensemaking loops involving the actors in the Mälarporten project, near- and distant future prospective sensemaking loops, characterized by different steps as illustrated in Figure 7. The near-future loop closely relates to the development process of the AI-application, while the distant-future loop relates to the complex systems development process, i.e. the energy system in the urban development process. Although both loops depend on active participation from many actors, one or the other seem to be put in the foreground depending on current project activities and the inclinations of participating actors. The interrelations between the loops, indicated with dotted arrows in Figure 7, becomes important cues to understand the nuances of prospective sensemaking processes in complex system development.

While the findings confirm that an AI-application can augment human cognition by increasing the information basis for decision-making in the urban planning process, they also indicate a potential impact on the organizational level, altering timing and tasks in the decision-making process. Together the two loops give indications on how narrowing down the imagination of a visionary distant-future towards more specific changes and applications in the near-future of the urban planning process can take place. In this process, the development of the AI-application can serve as a catalyzer to bridge the gap between distant- and near-future imagination. It also suggests how the past in form of historical data and human experiences are connected to desired futures in the present.

The paper adds to existing literature on sensemaking and desired futures by further explicating the process of connecting the distant-future imagination to the near-future imagination and identifying important steps towards actualize desired futures. It

widens the understanding of sensemaking in the transition towards complex intelligent systems by highlighting how past, present, and future (near- and distant) need to be connected in the development phase.

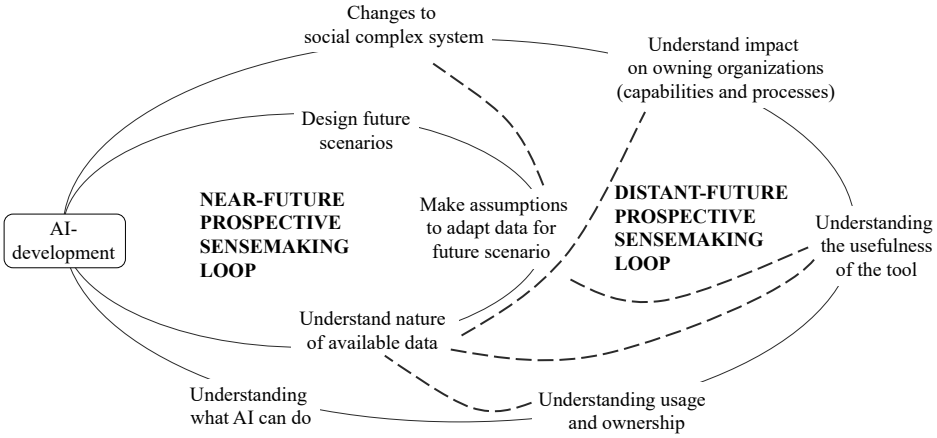


Figure 7. Interplay between near-future and distant-future sensemaking loops

5.2.2 Contribution to thesis

This paper contributes to the thesis by adding insights on the research question "*What role can AI-solutions play for prospective sensemaking in the development of complex system?"*. The paper contributes with insights based mainly on the project level and complex system level of analysis, but also to some extent to the social system level. By identifying the near- and distant-future sensemaking loops in an AI-development project within complex systems, the paper sheds light on the role of AI in prospective sensemaking. The interrelationships between the loops indicate tight dependencies between them and gives insight to how they influence each other. The interdependencies also indicates how AI cannot only augment human cognition but also augment organizational capabilities by e.g. adapting work processes.

6 Discussion

The aim of this thesis is to *explore collective prospective sensemaking processes in the development of future complex intelligent systems* and to do so, I have drawn on the two appended papers addressing the research questions:

> *RQ1. How are prospective collective sensemaking processes enacted in the FEI of an AI-project?"*
>
> *RQ2. What role can AI-solutions play for prospective sensemaking in the development of complex system development?*

6.1 A tentative framework for prospective sensemaking in future complex intelligent systems

To explore the prospective sensemaking as AI is introduced in complex systems, I examined dimensions of prospective sensemaking in relation to important characteristics of complex systems development. Consideration of type of complexity in complex system development can give suggestions for actions. By presenting a *future-complexity framework* to analyze near- and distant-future prospective sensemaking in relation to internal and external complexity, new insights to complex system development emerge. I have developed the framework building on the findings and conclusions arrived at in the papers; near- and distant-future prospective sensemaking being used on one axis, and the sensemaking behaviors mapped into each box in the framework. Combined with types of complexity on the other axis, this lays the foundation for the analysis of system levels and knowledge flows that seem important for the sensemaking processes.

Misalignment in the sensemaking processes between domain actors and AI-developers was discussed in paper 1, identified sensemaking behaviors: seeking, disengaging, and supporting. Temporal misalignment in sensemaking needs and the enactment between the actors was highlighted. This emphasized aspects relating to the prospective collective sensemaking process between groups with different knowledge bases and backgrounds involved in AI development for complex systems that add to previous works by e.g. Stigliani and Ravasi (2012) and Gattringer et al. (2021). Paper 2 shows differences in prospective focus in the sensemaking process between the groups and how both distant-future prospective sensemaking and near-future prospective sensemaking contribute to the understanding of how to take action (Rindova & Martins, 2022). Both focuses are important in the development of complex intelligent systems and enhance the understanding of the sensemaking processes within the complex systems literature, nuancing what constitutes collective sensemaking as being something more than multiple, simultaneously ongoing, sensemaking processes.

In complex systems development, uncertainties and unknowns are significant (Davies & Hobday, 2005) and actors in complex systems are used to deal with them. Organizations adapt by defining and standardizing processes and other mechanisms to detect and take measures to mediate emerging behaviors. Such uncertainties can be related to internal complexities as defined by Rousseau (2019). In the development process of complex systems, external complexity (Rousseau, 2019) is prominent as reflected by the need for system thinking (Senge, 1997) and sensemaking (Weick, 1995) which are well present in the CoPS literature. The AI-tool in the studied case is an attempt to use AI to deal with some of the external uncertainties in the urban planning process. However, the internal complexity also needs consideration to realize such a tool.

In Figure 8, I present the main activities performed within each box in the four-field framework. The framework explicates actor behaviors within the case project to further enhance our understanding of actor relations and the influence on prospective collective sensemaking processes within complex intelligent system development. I elaborate each box of the framework below which lays the basis for the analysis of knowledge flows across system levels and boundary objects in the next section. Each box is given a title drawn from the quotes in the empiric study.

Distant-future prospective sensemaking	1. Planned and desired future scenario details, adaption of data. Validation of models. Domain actors support sensemaking, disengage from collective sensemaking. AI-developers seeking collective sensemaking	2. Desired future scenarios and consequences of the AI. Domain actors seeking collective sensemaking. AI-developers support sensemaking, disengaged from collective sensemaking
Near-future prospective sensemaking	3. Historical data, access and quality. Domain actors disengaging collective sensemaking. AI-developers disengaging collective sensemaking	4. Urban planning process, rules and regulations. Domain actors support sensemaking, disengaged from collective sensemaking. AI-developers seeking collective sensemaking
	Internal complexity	***External complexity***

Figure 8. Explicating types of prospective sensemaking in relation to complexity characteristics

1. Tell me what the future needs are, and I'll give you a model to predict the future

When intersecting *distant future* and *internal complexity,* the focus of the team is on understanding the city plan and what desired future scenarios would look like to adapt available data and build models to match these future scenarios. The domain actors act mainly to support sensemaking for the AI-developers, whereas AI-developers tend to seek collective sensemaking to understand distant future requirements. This was demonstrated in paper 1 with a misalignment in sensemaking processes as domain actors had left the project meetings and thereby disengaged form collective sensemaking. Making sense of distant future and internal complexity implies pinning down the distant-future to be useful for advancing the AI-development operating on internal complexity, i.e. to promote action in the present.

2. I understand that AI is the future, I just don't understand what the AI developers are doing

When intersecting *distant future* and *external complexity,* the focus is on defining desired future scenarios and understand consequences and possibilities of an AI introduction. The domain actors then seek collective sensemaking to understand what AI can do in their field specifically while the AI-developers support sensemaking for domain actors while focusing on what they need to start the technical development and thus disengage collective sensemaking. This is also demonstrated by the misalignment of sensemaking processes in paper 1, when AI-developers started the development and while the domain actors continued the discussions around usage and ownership. Sensemaking of distant future and external complexity seem to be less connected to the AI-project at hand, but rather the possibilities of AI in general, disconnecting it from the internal complexity.

3. I just have one data source, not even ten, just one

When intersecting *near future* and *internal complexity* focus is on accessing data and understand the quality. The domain actors found it hard to understand the requests for specific data that the AI-developers made, whereas AI-developers focused on developing their modules with the data that was available. Neither part was seeking collective sensemaking. Sensemaking of near future and internal complexity seem very focused on the AI-project at hand and tend to get disconnected from the external complexity and distant-future.

4. Can't the municipality just buy the tool?

When intersecting *near future* and *external complexity* focus is on understanding how the urban planning process works and what rules and regulations need to be considered. The domain actors act mainly to support sensemaking for the AI-developers, whereas AI-developers tend to seek collective sensemaking to understand the context. This was demonstrated in paper 1 with a misalignment in sensemaking processes as domain actors had left the project meetings and thereby disengaged form collective sensemaking. Making sense of near-future and external complexity focuses on practical constrains and conditions that could influence work in the present.

From the framework internal complexity seems tighter connected to near-future sensemaking as the actors disengage from collective sensemaking. Similarly, external-complexity seems more connected to distant-future sensemaking. In the distant-future external (2) and near-future internal (3) boxes, collective sensemaking is less likely to occur spontaneously than in the distant-future internal (1) and near-future external (4) collective sensemaking where collective sensemaking seems to be triggered by the exposure of knowledge as it is sought by some actors. Prospective collective sensemaking thus seem to need more support in 2 & 3, alternatively to encourage activities in 1 & 4 could cause spillover effect on 2 & 3. Alternating activities between 1 & 4 and 2 & 3 could be beneficial to stimulating continuous prospective collective sensemaking.

6.2 Near- and distant-future sensemaking

The framework also show how the sensemaking approaches identified in paper 1 relate to near- and distant prospective sensemaking in paper 2. Distant-future prospective collective sensemaking can be connected to seeking approaches by domain actors taking place in the beginning of the project when the overall project focus is on goals and how to use the AI-tool. Correspondingly, AI-developers seek distant-future prospective collective sensemaking later in the project when the focus is on building the tool. Near-distant prospective collective sensemaking is sought by AI-developers whereas domain actors do not actively seek it. However, both near- and distant future sensemaking are present throughout the development process and most actors need to engage in both near- and distant-future sensemaking. The interplay between the two types of prospective sensemaking processes are important factors for managing the collective prospective sensemaking process in complex intelligent systems as demonstrated in paper 2, where seeking and disengaging behaviors from paper 1 can give key insights. The framework adds to these findings by identifying the conditions in box 1 & 4 to stimulate collective sensemaking. The introduction of AI seems to reinforce the need for simultaneous consideration of near- and distant-future perspectives, as it precipitates the need to adapt the historical data for futuristic use. This can increase the importance of conscious consideration for the involvement of

actors in different phases and the situated epistemic uncertainty experienced (Walker et al., 2003) in complex system development. The insights advance the understanding of the collective prospective sensemaking process (Stigliani & Ravasi, 2012) in the context of cross discipline collaborations (Gattringer et al., 2021) by highlighting which behavior different actors show depending on near- or distant-future sensemaking focus.

6.3 Internal and external complexity

The creation of the AI-tool emphasizes the interrelation between external and internal complexities, stressing the need for collaboration between different tasks and roles (Ozkaya, 2020) and gaps in knowledge need to be bridged (Piorkowski et al., 2021). Furthermore, distinguishing complexities as internal or external can nuance the understanding of how AI affect uncertainties in complex systems (Benbya et al., 2021; Berente et al., 2021; Ozkaya, 2020; Sagodi et al., 2023). The AI-developers do not seem to seek collective prospective sensemaking for coping with internal complexity, while domain actors seek it mostly to understand the consequences on a distant-future time horizon. For coping with external complexity, AI-developers seek near-future collective prospective sensemaking, while domain actors seek distant-future sensemaking, i.e. the time horizon that is most relevant for their task and responsibility. AI introduction seems to possibly increase internal complexity, while the aim in the empirical case studied related to decreasing external complexity with the help of AI. As mentioned in the theoretical background, external complexity requires holistic methodology like system thinking, whereas internal complexity requires reductionist methodology dividing the system in manageable pieces (Rousseau, 2019). The team was observed to struggle to balance between these two approaches in the case study, which contributed to the misalignment of sensemaking processes in paper 1. The focus differed between the actors and understanding for both internal and external complexity was challenging as reflected in paper 2.

Additionally in complex systems development, prospection becomes prominent to handle potential behaviors in the future (Jacobs et al., 2013) in contrast to handling emerging behaviors in the present. The study indicates AI's role in connecting retrospective sensemaking with prospective sensemaking by adapting historical data to visions of probable and desired futures. This entails connecting the internal complexity to external complexity in which AI can be beneficial. The process reveals possible paths and prime considerations and actions to approach desired futures that seemingly can decrease the experienced distance to desired futures for the involved actors and stakeholders (Gümüsay & Reinecke, 2022). When future scenarios are simulated with the help of AI technologies, they become explicit and easier to imagine and thus to work on paths to realize them. Although the scenarios might not represent the final goal of a desired future, the feasibility is easier to evaluate as well as if it is in line with a desired future or would have undesired consequences.

6.4 Knowledge flows across system levels

To further explore prospective collective sensemaking in the emergence of complex intelligent systems, I have explicated the knowledge flows for each box in the framework to give further insights on what is important to consider when AI is introduced into complex systems development.

Figure 9 demonstrates that with a distant-internal focus (box 1), the knowledge trickles down from social system level, to complex system level, to project level in the process of understanding the planned city district and future scenarios. However, box 1 also encompasses the actors seeing how the social system organization could be adapted to make better decisions for the future, in the future, i.e. a flow from project level, to complex system, to social system level. With a distant-external focus (box 2), knowledge flows in both directions between social system level and complex system level, visible in vivid discussions around usage and ownership. However, knowledge also flows from project level to complex system level to social system level to understand the possibilities and limitations of AI applications to apply it to the context of energy systems. With a near-internal focus (box 3) no knowledge really flows over the system levels which can be understood as the knowledge needed for finding data and evaluating the quality on the project level is a straightforward task not really requiring collective effort and knowledge input from other levels. Likewise, the domain actors did not perceive that the activity would influence the complex system level and social system level at this point. With a near-external focus (box 4) knowledge flows from social system level and complex system level to project level as the urban planning process needs to be understood and adhered to.

How the knowledge flows between system levels for the different prospective sensemaking objectives and complexity focus, enhances the understanding of *collective prospective sensemaking processes in the development of complex intelligent systems* and give further insights on what is important to consider when AI is introduced into such systems.

With regards to distant-future sensemaking, iterating knowledge between levels seem to become more important which is demonstrated by a reciprocal knowledge flow for both internal and external complexity understanding. More static or one-directional knowledge flows appear present in near-future sensemaking. This indicates how AI applications can influence the organization on complex system level and social system level, where new knowledge can influence the urban planning process. The direction of the knowledge flow identified in the framework gives an indication to when it could be beneficial to use holistic or reductionist methodologies to face challenges like those identified in the empirical study (Rousseau, 2019). Correspondingly it can give insights

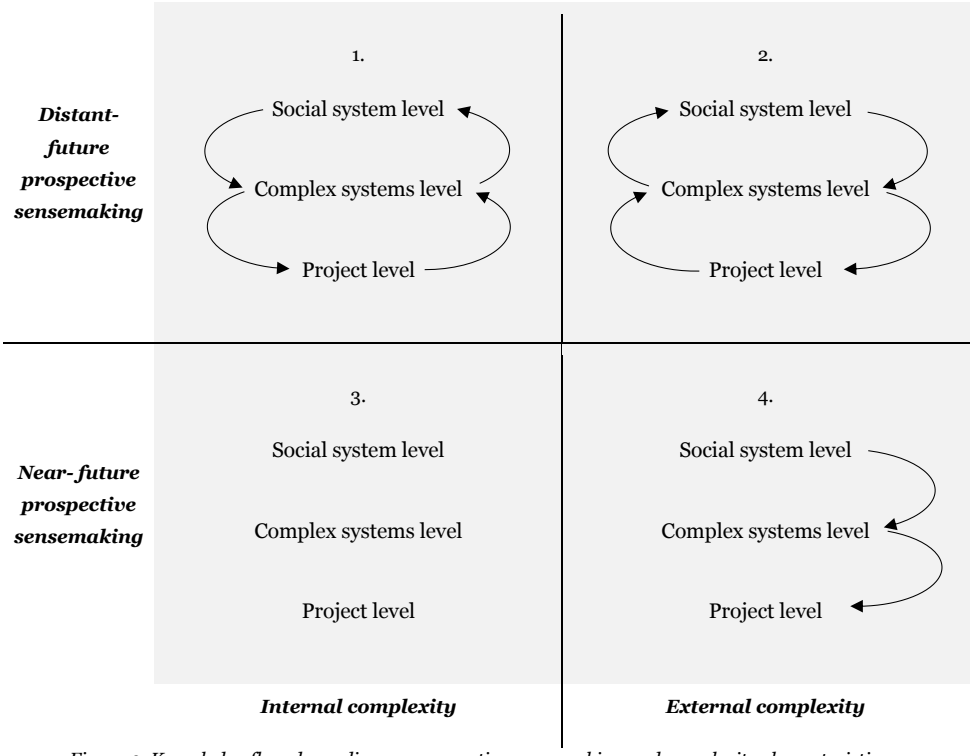


Figure 9. Knowledge flow depending on prospective sensemaking and complexity characteristics

on how it is possible to balance between them depending on the type of complexity in focus and on which system level the receiver is situated. Knowledge flows towards social system level could be argued to require holistic methodologies and knowledge flows towards project level would be favored by reductionist methodologies and thus indicate in what situations in the development process one or the other could be beneficial. For distant-future prospective sensemaking both holistic and reductionist methodologies seem to be needed to support knowledge flows in both directions, while near-future prospective sensemaking would require primarily reductionist methodology.

Considering the knowledge flow over system levels in prospective sensemaking gives a perspective that can aid the collective process when diverse actors with knowledge and focus on different system levels need to collaborate, as is the case with AI introduction in complex systems. This advances our understanding of the prospective collective sensemaking process with diverse actors (Gattringer et al., 2021). It adds to the knowledge of the role of facilitators as it advances the understanding of the sensemaking processes and give insights to how the prospective collective can be directed towards internal or external complexity. Facilitating the knowledge flow in the

'right' direction can also decrease the epistemic uncertainty experienced by many actors in relation to AI-development, i.e. caused by imperfections in our knowledge (Walker et al., 2003).

6.5 Boundary objects

The framework also provides insights around the use of boundary objects discussed in paper 1 (Carlile, 2002; Star & Griesemer, 2015). It appears important that the boundary objects support the type of complexity that is in focus in the project meetings. Boundary objects with a high level of detail primarily support the near-distant future sensemaking. However, such boundary objects also benefit the distant-future sensemaking when questions spurred by near-future considerations provoke distant-future sensemaking. External complexity does not seem to call for boundary objects with high level of detail, but it can be questioned if too high level of detail would draw the focus toward the internal complexity and thus not be favorable to support sensemaking around the external complexity for neither near- nor distant-futures. The framework also relates supporting behavior identified in paper 1, assisted by boundary objects (Carlile, 2002; Star & Griesemer, 2015), with the interrelation between near- and distant-future sensemaking loops from paper 2 which seem important for the knowledge flow between system levels. The level of detail of the boundary objects could be a tool for project management to direct the prospective collective sensemaking process towards internal or external complexity and thus direct the knowledge flow depending on the need in the phase that the project is in. The tentative framework advances our understanding of boundary objects and how to use them depending on the detail level. An area that has been raised to need further research by Majchrzak et al. (2012). Adapting the level of detail in boundary objects also resonates with Rousseau (2019) to balance reductionist and holistic methodologies in general.

7 Conclusion and Future Research

7.1 Conclusion

This thesis presents an empirical case of AI-development in a complex system, an important contribution for the evolving phenomenon of AI introduction in complex systems that gives insights on the transformation into complex intelligent systems. Furthermore, the focus on the development process of complex systems, adds a clear prospective perspective that has not previously been studied explicitly to any larger extent. By examining the prospective collective sensemaking processes present in the development process of future systems, this thesis highlights temporal aspects of perspectives on the future. The proposed future-complexity framework combining near- and distant-future perspectives and internal and external complexities suggests that collective sensemaking needs more support for near-future internal and distant-future external phases. It shows a connection between type of complexity and the required detail level of boundary objects used to support prospective collective sensemaking in such phases. The new insights advance our understanding on different aspects of AI introduction in complex system development that influence the prospective collective sensemaking process. Insights that give better opportunities for management of the sensemaking process. The thesis highlights the connection between retrospective and prospective perspectives that become reinforced in the development of complex systems when AI is introduced, as we turn to historical data to assist in designing the future. The framework also discusses the influence of knowledge flows between different system levels in the AI introduction in complex system development and how this can increase our awareness in methodological considerations and the use of boundary objects.

7.2 Managerial implications

For managers this thesis gives useful knowledge about different aspects influencing the prospective collective sensemaking process summarized below. These can be applied to planning and adapting the development process to address the different sensemaking needs and different knowledge backgrounds of the participating actors that seem important when introducing AI in future complex systems.

- Acknowledge the necessity for both near- and distant-future sensemaking and facilitate ways to shift between them and to explore connections. Recognize that AI introduction will tie retrospective and prospective sensemaking tighter together and support the process of understanding how retrospective data will support prospective sensemaking augmented by AI.

- Consider that prospective collective sensemaking is more easily triggered in near-future external complexity conditions and distant-future internal complexity conditions. Alternate the focus between these and near-future internal and distant-future external conditions to stimulate prospective collective sensemaking. Be prepared to support the prospective collective sensemaking near-future internal complexity and distant-future external complexity conditions.

- Recognize at which system level different actors have their main interest and knowledge to understand when each actor is likely to seek, disengage, and support collective sensemaking. Try to meet the sensemaking need of the actors and consider the knowledge flow between system levels to facilitate the prospective collective sensemaking process.

- Recognize the nature of the complexities and the future time horizon in the issues raised in meetings to support the prospective collective sensemaking process using boundary objects of adequate level of detail.

7.3 Limitations

The thesis builds on a single qualitative case study which could limit the transferability to more generalizable theoretical conclusions. However, aiming for analytical generalizability from a single case study, carefully selected for the research topic at hand, can still prove valid (Flyvbjerg, 2004). Nevertheless, to further advance generalization to a broader context, more studies of AI introduction into complex systems, in energy systems and urban planning may make the findings and conclusions more dependable. Studies in other types of complex systems would create a more empirically varied foundation for theorizing (Eisenhardt & Graebner, 2007).

The empirical study that this thesis builds on was initiated when the AI introduction project had already been ongoing for approximately one year. This gives limited insight to the sensemaking processes in the early phases of the project. However, some information about this period and episodes that occurred before the start of the empirical study have been accessed through interviews and documents.

7.4 Future research

Building on the limitations, one path for the continued research for the doctoral thesis is to increase the dependability by exploring another case of AI introduction in complex systems. This could be within urban planning and energy systems or a related infrastructure system, or it could be within another sector such as health care. A second case study in the same kind of social complex system would give more in depth understanding and add dependability to findings and conclusions of this thesis. It

would also benefit from the general understanding of the social complex system that I have already assimilated. However, a case in another social complex system context would provide findings that can strengthen the transferability of the findings and conclusions but would require more effort to understand the new context of the complex system.

Another path forward could be to extend the theoretical focus in the direction of knowledge creation. This field has emerged during the data analysis as a field that would complement the prospective sensemaking perspective and could extend the understanding of prospective collective sensemaking processes in complex system development. E.g. examining how knowledge is created in relation to the knowledge flows across system levels when introducing AI in complex system, or how AI influences the creation of knowledge on different system levels in complex systems development e.g. relating to epistemic and variability uncertainty.

A third possible future path would be to make a continued study of the empirical case presented in this thesis where the implications of the AI project and the AI-tool on the organizational level could be followed over time. This would probably imply a greater focus on business model development which was mentioned on many occasions during the data collection as an important next step to reap the benefit from the AI-tool developed. Anticipating the shifts needed in future business models in energy systems is also seen as a big challenge with the diversity of actors involved in complex system development.

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Appended papers

Paper 1:

Särner, E., Yström, A., Lakemond, N., & Holmberg, G., (2023) *Prospective sensemaking in the FEI of AI-projects* [Accepted for publication in Research-Technology Management]

Paper 2:

Särner, E., Yström, A., Lakemond, N., & Holmberg, G., (2023) *Utilizing AI in prospective sensemaking for desired futures: Outlining near- and distant-future sensemaking of complex system development* [Unpublished manuscript]

Papers

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